

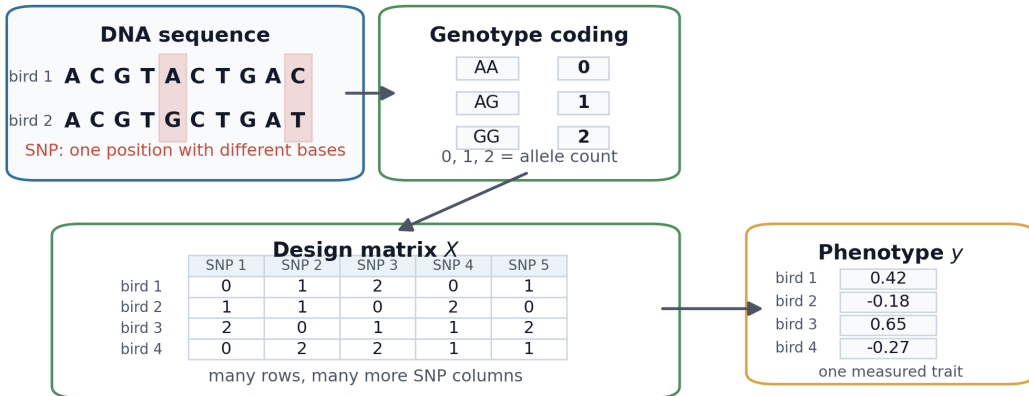
Training set optimization for genomic prediction across island populations

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SNPs and phenotypes



	SNP ₁	SNP ₂	SNP ₃	...	SNP _p
bird 1	0	1	2	...	0
bird 2	1	1	0	...	2
bird 3	2	0	1	...	1
⋮	⋮	⋮	⋮		⋮

$X_{ij} \in \{0, 1, 2\}$, $y_i = \text{measured trait}$.

The prediction problem is ordinary regression in an unusual regime: very many marker columns, fewer phenotyped individuals, and correlated columns.

Marker-based regression

$$y = \mu 1 + X\beta + \epsilon.$$

Each β_j is the fitted effect of one SNP column.

Mixed-model view

$$y_i = \mu + g_i + \epsilon_i, \quad g_i = x_i^\top \beta.$$

The genetic part g_i is latent, but the SNPs give us a high-dimensional proxy for it.

$$\hat{\beta} = \arg \min_{\beta} \{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \}.$$

Prediction engines: ridge and BPCRR

Ridge

Fit all marker effects with a common shrinkage penalty.

$$\hat{y}_* = \hat{\mu} + x_*^\top \hat{\beta}.$$

It is simple, stable, and a strong baseline in high dimensions.

BPCRR

Compress SNPs to principal components:

$$Z = XU_q, \quad q \ll p.$$

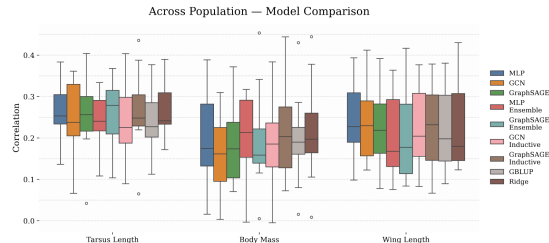
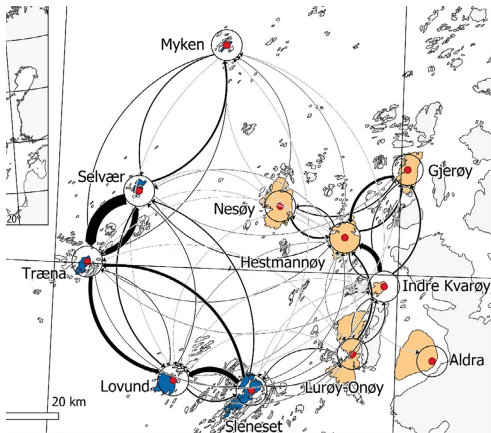
Then fit a Bayesian ridge model in PC space:

$$y = \mu 1 + Z\gamma + \varepsilon.$$

Takeaway

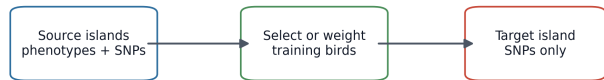
Both are regularized linear prediction models; they differ mainly in whether shrinkage happens in marker space or PC space.

When genomic prediction gets hard



- ▶ Within-population validation is close to ordinary supervised learning.
- ▶ Across-population prediction is a distribution-shift problem.
- ▶ In my data, the island structure gives a natural leave-one-island-out test.

Training set optimization



S : candidate training set

max predictive accuracy
under target-aware choice

T : birds we want
to predict well

Goal

Predict a target island T using source islands with phenotypes and SNPs.

Choice

Choose a subset S_k or weights w_i before fitting ridge/BPCRR.

Constraint

In deployment, X_T is available but the target phenotypes y_T are not.

target-aware training: $\max_{S_k \text{ or } w} \text{cor}\{y_T, \hat{y}_T\}.$

Three routes I tried

Method	Idea	What information it uses
Data Shapley	Estimate each source island's marginal contribution to target prediction.	Genotypes, source phenotypes, and calibration target phenotypes.
Top- k selection	Rank source candidates by genetic closeness or expected prediction uncertainty.	Genotypes and target SNPs.
Importance weighting	Reweight source data to make the training distribution look more target-like.	Genotypes and target SNPs.

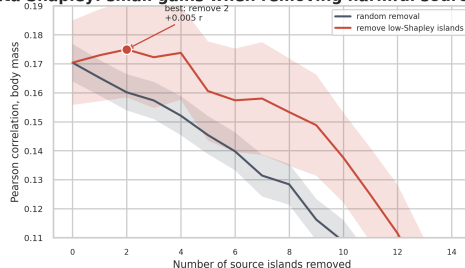
All results are evaluated with leave-one-island-out prediction and Pearson correlation on the held-out island.

$$\phi_i = \mathbb{E}_{S \subseteq D \setminus \{i\}} [V(S \cup \{i\}) - V(S)].$$

- ▶ $V(S)$: prediction performance when training on subset S .
- ▶ I used island-wise TMC Shapley to make it computationally feasible.
- ▶ Useful diagnostically, but calibration target phenotypes make it partly oracle-like.

Data Shapley: Ghorbani and Zou (2019).

Data Shapley: small gains when removing harmful source islands



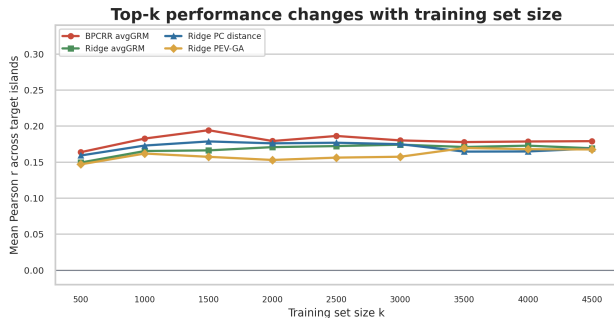
Top- k selection

Rank source candidate c by average genetic similarity to the target:

$$s(c) = \frac{1}{|T|} \sum_{t \in T} G_{tc}$$

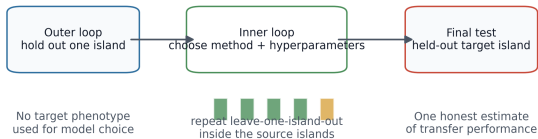
or in PC space:

$$d(c) = \|z_c - \bar{z}_T\|_2.$$



There may be a useful training-set size, but the best k is difficult to know a priori and differs by model.

Nested CV for target-aware rules



Nested CV prevents tuning the training-set rule directly on the island we later report as test performance.

- ▶ Outer loop: hold out the target island.
- ▶ Inner loop: choose model, weighting rule, and hyperparameters using only source islands.
- ▶ Final: retrain on all source islands and evaluate once on the target island.

Takeaway

This keeps the reported target performance honest.

Target risk by source weighting

Under covariate shift,

$$R_T(f) = \mathbb{E}_T\{\ell(f(X), Y)\}$$

$$R_T(f) \approx \mathbb{E}_S\{w(X)\ell(f(X), Y)\}.$$

where

$$w(x) = \frac{p_T(x)}{p_S(x)}.$$

In practice I estimate this with logistic classification in PC space, then clip/smooth weights for stability.

Bayes ratio from a domain classifier

Let $D \in \{T, S\}$ indicate target or source.

$$p(D = T \mid x) = \frac{p(x \mid D = T)p(D = T)}{p(x)}.$$

Therefore

$$\frac{p(x \mid D = T)}{p(x \mid D = S)} = \frac{p(D = T \mid x)}{p(D = S \mid x)} \frac{p(D = S)}{p(D = T)}.$$

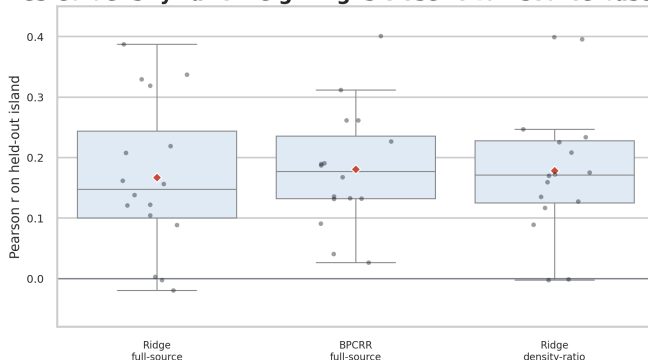
Importance weighting results

- ▶ Density-ratio weighting was tuned inside nested CV.
- ▶ The weighted models are close to full-source baselines.
- ▶ This is more conservative than top- k , but has not clearly improved prediction yet.

Takeaway

Target-aware weighting has not yet given a robust win under strict validation.

Nested density-ratio weighting is close to full-source baselines



What seems to work

- ▶ Genetic similarity contains signal for training selection.
- ▶ Top- k BPCRR/avgGRM gives the clearest preliminary improvement.
- ▶ Shapley is useful for diagnosing harmful source islands.

What is hard

- ▶ The best k is hard to know without target phenotypes.
- ▶ Shapley is expensive and calibration-dependent.
- ▶ Weighting can reduce bias, but may increase variance or effective sample-size problems.

Takeaway

Future work: better a priori k -choice, stability diagnostics for weights, and clearer separation of true transfer gain from island-specific noise.

Thank you

Questions?